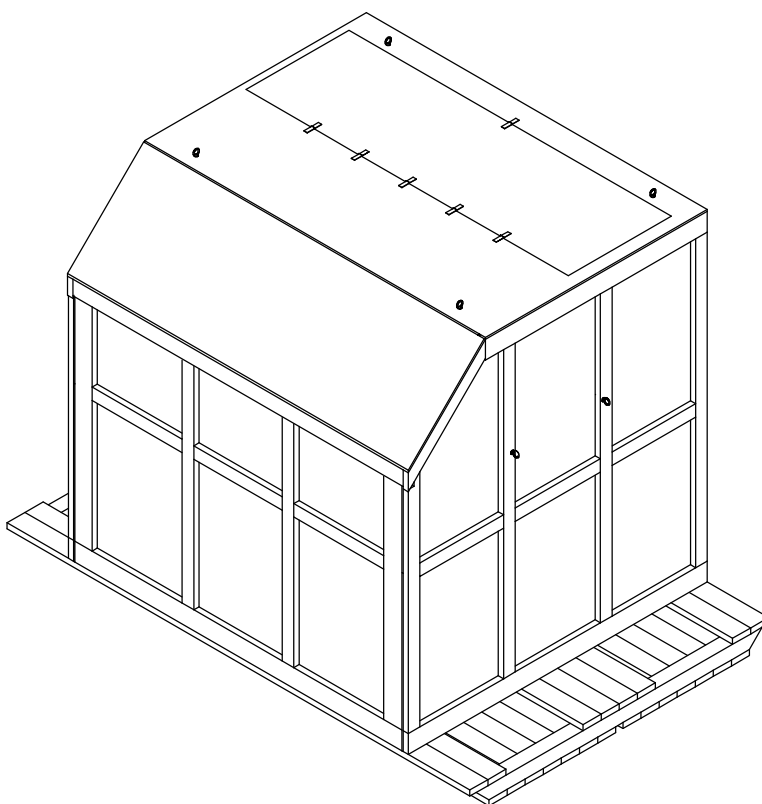


SECTION 3

INTERNATIONAL (AIR / SEA) MAGNET DELIVERY

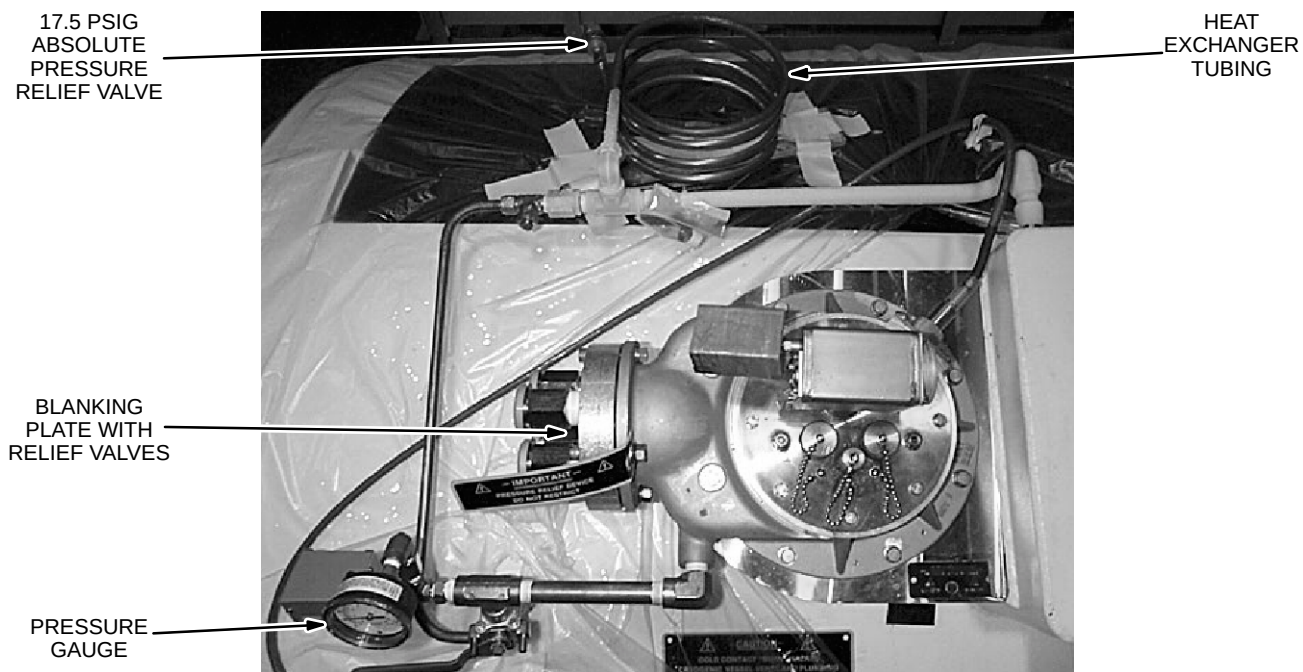
Note

The following information outlines provisions for filling and electrical / temperature checks while the magnet is still in the shipping crate. This will enable required service to be performed in transit based upon magnet shipping date.

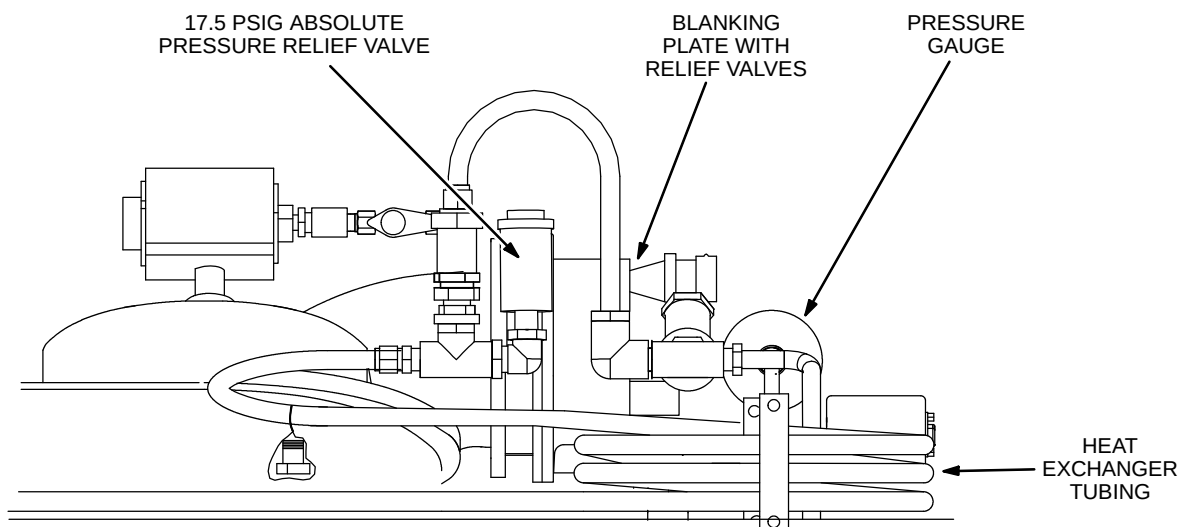


CRATE / PALLET
ILLUSTRATION 3-1

3-1 IN-TRANSIT HANDLING AND SERVICE



MAGNET ASSEMBLY 2188846



MAGNET ASSEMBLY 2243832

TOP ACCESS DOOR
ILLUSTRATION 3-2

3-1 IN-TRANSIT HANDLING AND SERVICE (continued)

CAUTION

Review all shipping and handling guidelines in Tables 3-1, 3-2 and 3-3 with the Shipper / Riggers prior to transporting the magnet.

These guidelines must be followed to prevent any potential damage to the magnet during shipping and handling.

TABLE 3-1
IN-TRANSIT LIFTING / HANDLING INFORMATION

MAGNET	MAXIMUM WEIGHT ^{a,b}	MAXIMUM TILT ^c	FORKLIFT CAPABILITY
LCC 1.5T	14,700 pounds (6,670 kg)	30 degrees	Yes
LCC 1.0T			
<			

CAUTION

The magnet CANNOT be shipped by train as serious damage to the magnet's internal suspension system may occur due to the vibration loads introduced by rail systems.

TABLE 3-2
SHIPPING INFORMATION

MAGNET	ALLOWABLE SHIPPING MODES	SHIPPING CAPABILITY	MAXIMUM TRANSIT TIME
LCC 1.5T	Airplane, air ride only trailer, boat or ocean-going ship	Warm or Cold	Cryogen refill for cold shipment = 15 days
LCC 1.0T			

3-1 IN-TRANSIT HANDLING AND SERVICE (continued)

TABLE 3-3
MAXIMUM LOAD INFORMATION

MAGNET	MAXIMUM LONGITUDINAL LOAD	MAXIMUM LATERAL LOAD	MAXIMUM VERTICAL LOAD	MAXIMUM SHOCK LOADS
LCC 1.5T	1.2 G	1.5 G	1.5 G	None Allowed
LCC 1.0T				

3-2 ADDITIONAL EQUIPMENT / TOOLS FOR CRATED MAGNET INSTALLATION

3-2-1 Forklift / Crane

TABLE 3-4
CRATED MAGNET LIFTING / HANDLING INFORMATION

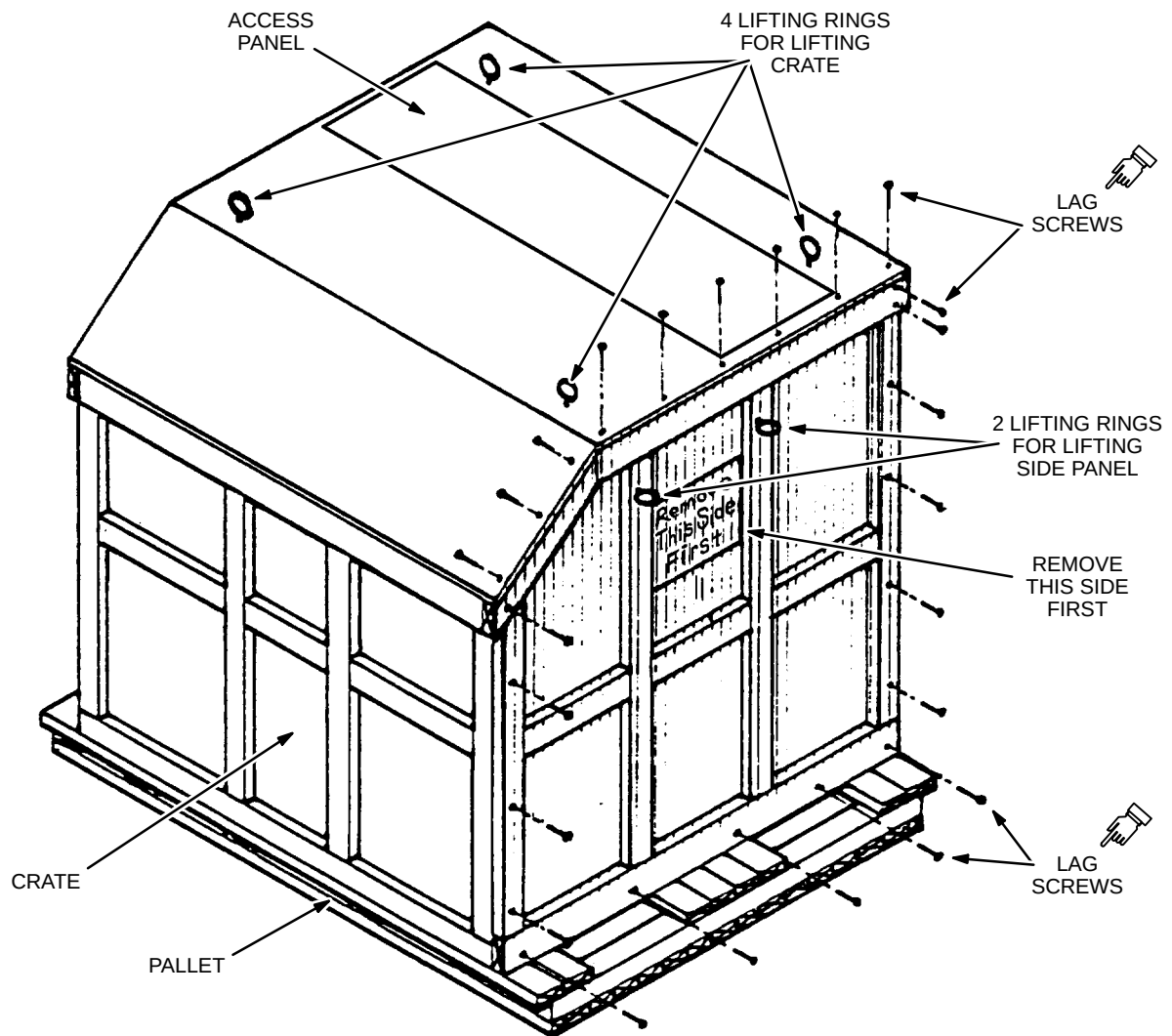
ITEM	QTY.	SPECIFICATIONS		FURNISHED BY	INTENDED USE
Forklift	1	Forklift Capacity:	15,000 pounds (6,804 kg) at 50 inches (1,270 mm) to load center	Rigger	Unloading or moving magnet with pallet and crate installed
		Minimum Fork Length:	90 inches (2,286 mm)		
		Minimum Distance Between Forks:	40 inches (1,016 mm)		
Crane	1	Total Load:	15,000 pounds (6,804 kg)	Rigger	Unloading or moving magnet with pallet and crate installed
Sling (strap or cable)	4	Work Load Limit:	1 ton (907 kg)	Rigger	Crate removal
		Minimum Length:	72 inches (1830 mm)		
Shackle or hook	4	Work Load Limit:	1 ton (907 kg)	Rigger	Crate removal
Wrench	1	Sizes:	9/16 and 3/8 inches	Rigger	Crate removal
Note: For handling and lifting magnet outside crate, refer to Section 2.					

3-3 CRATE REMOVAL

1. Remove all subsystem crates, except the magnet crate, from the low-boy trailer using a crane or forklift. Inspect all crates for visible damage.
2. Move subsystem crates to a receiving location protected from the weather, preferably in proximity of the examination room and on the same level.

3-3 CRATE REMOVAL (continued)

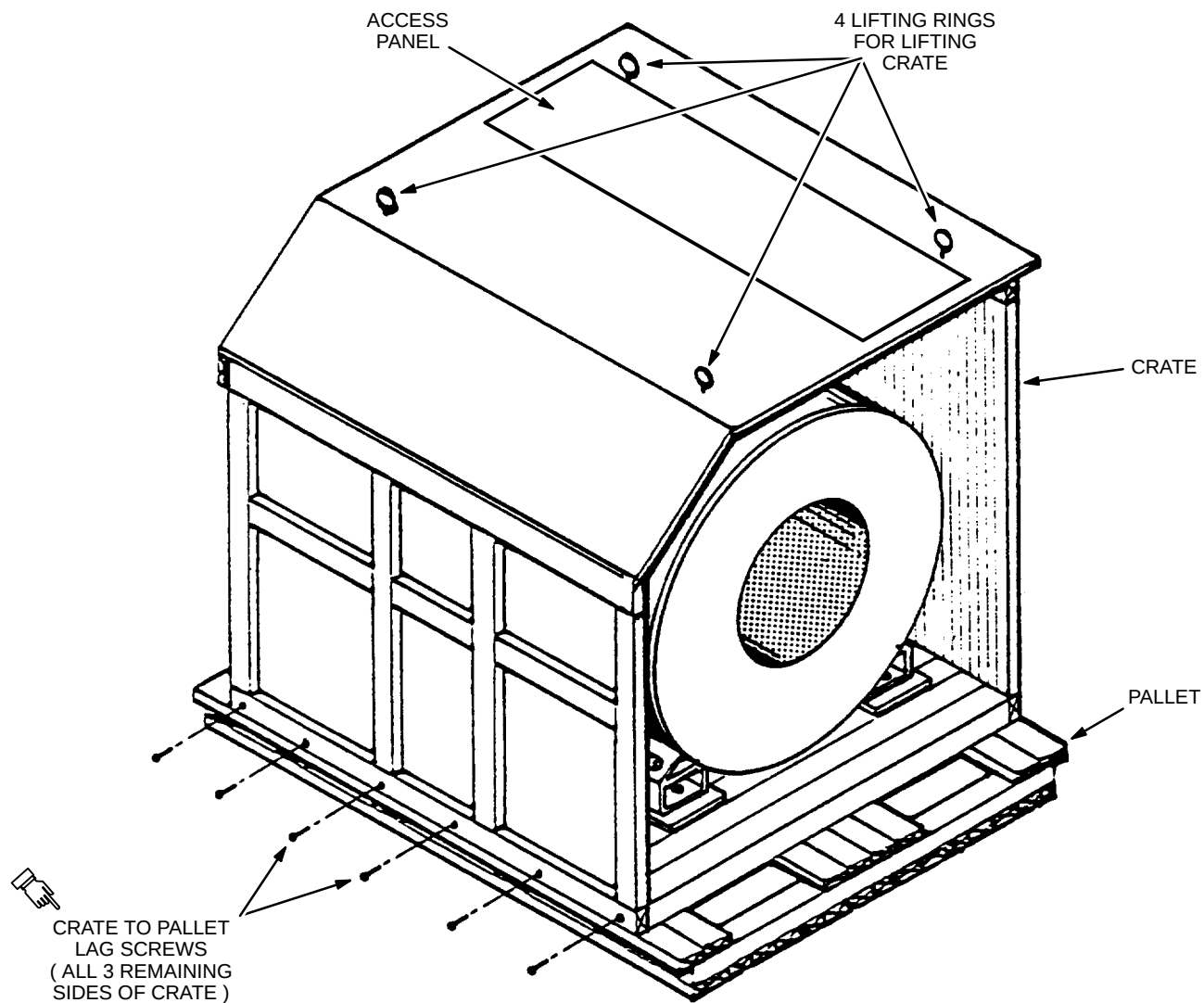
3. Inspect the magnet crate and identify the crate side marked, "REMOVE THIS SIDE FIRST," which must be removed before the others.
4. Position crane centrally over the 'Remove First' side of the crate.
5. Connect the crane to the two lifting rings located near the top of the 'Remove First' side of the crate. Use a sling and shackles supplied by riggers. Tighten the sling snug.
6. Remove the lag screws securing the 'Remove First' side to the rest of the crate. See Illustration 3-3.
7. Lift the 'Remove First' side of the crate with the crane and place the side in a clear area..



MAGNET CRATE FACE LAG SCREW REMOVAL
ILLUSTRATION 3-3

3-3 CRATE REMOVAL (continued)

8. Secure four one ton (907 kg) working load slings, minimum 6 foot length (1829 mm), to the four lifting rings assembled to the top corners of the crate. Using one ton (907 kg) anchor shackles.
9. Position the crane centrally over the top section of the crate and attach the slings to the crane hook.
10. Remove lag screws securing the three remaining crate sides to the pallet. See Illustration 3-4.



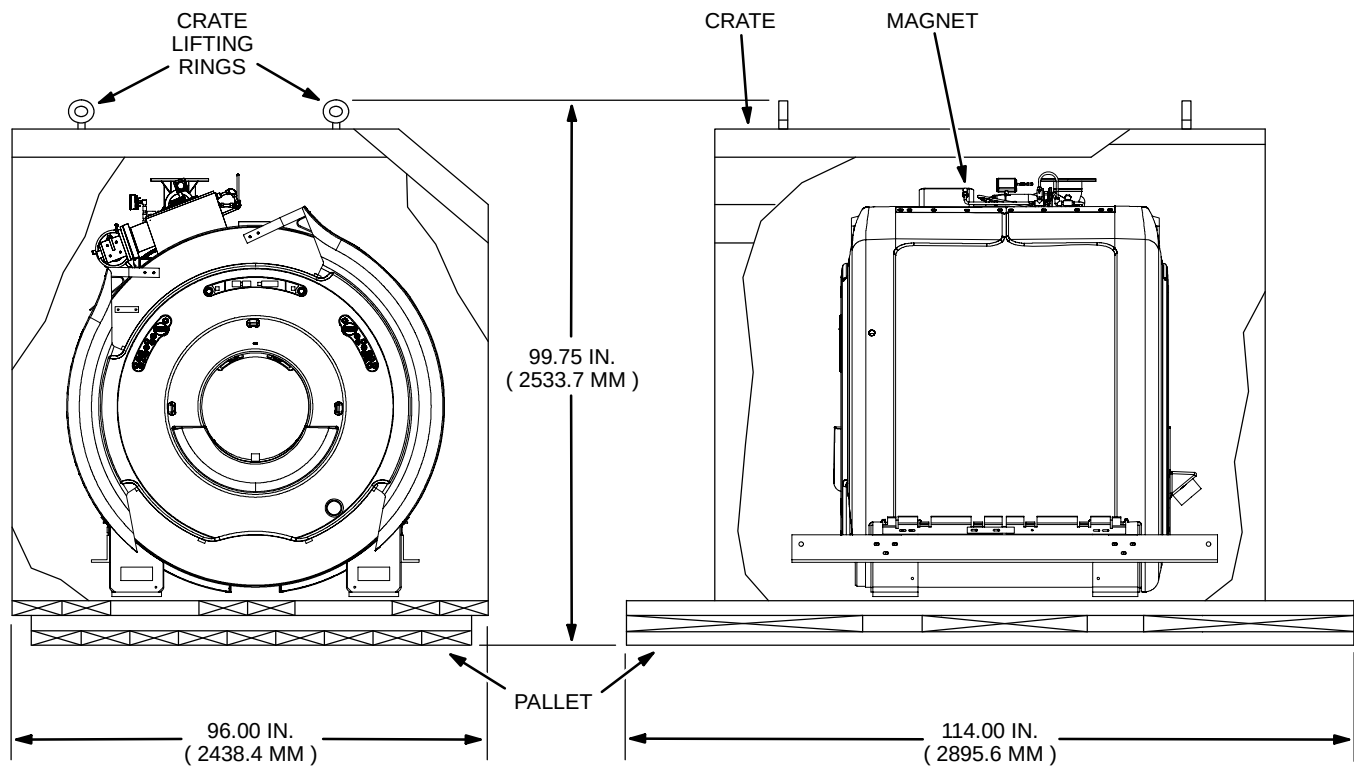
MAGNET CRATE BOTTOM LAG SCREW REMOVAL
ILLUSTRATION 3-4

3-3 CRATE REMOVAL (continued)

CAUTION

Care must be taken not to scrape or hit the sides of the magnet with the crate.

11. Carefully lift the crate using the crane and clear the crate from the magnet's shipping pallet.



MAGNET CRATE
ILLUSTRATION 3-5

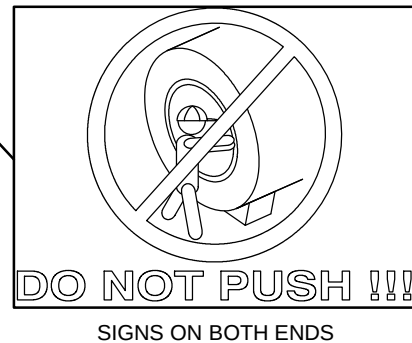
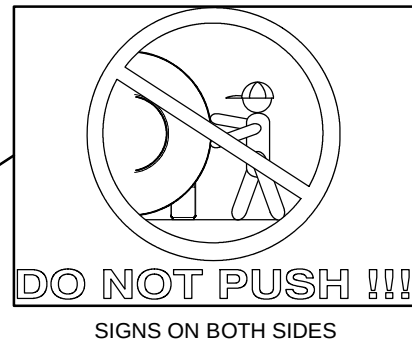
3-4 UNLOADING MAGNET AND PALLET

WARNING!

DO NOT APPLY ANY FORCE TO FIBERGLASS COVERS ON TOP, SIDES AND ENDS (SEE ILLUSTRATION 3-6).

USE LIFTING CABLES AS SHOWN TO LIFT MAGNET (REFER TO SECTION 2-2-2).

BEFORE MOVING THE MAGNET TO MR SUITE, REFER TO SECTION 2-5.



"DO NOT PUSH" SIGN LOCATIONS
ILLUSTRATION 3-6

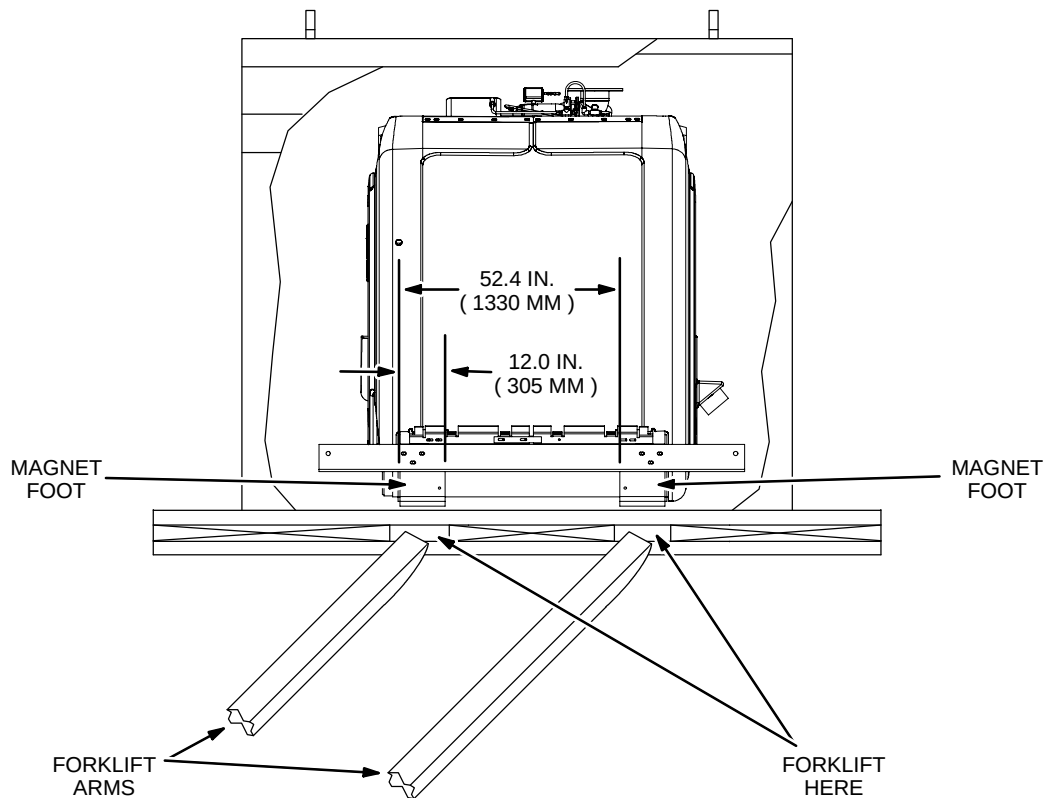
3-4-1 Forklift Method



Extreme care must be exercised during forklift operations:

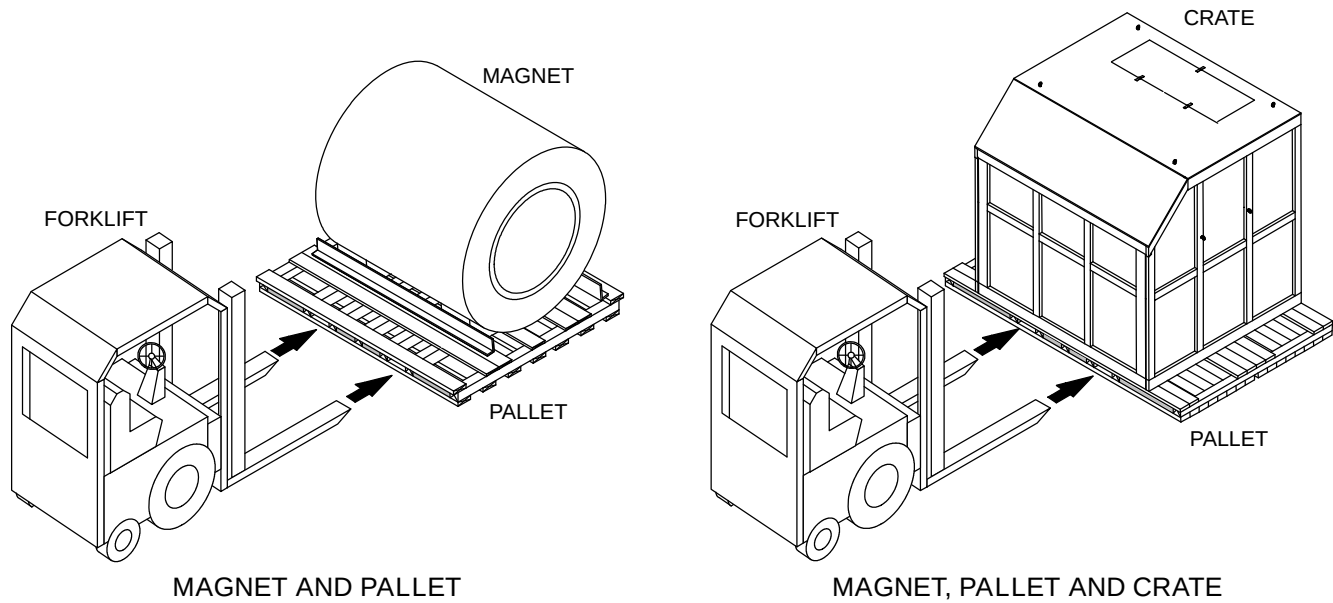
- S It is important to use the proper size forklift. Make sure the forklift capacity is rated for the weight of the crate and the contents. See Illustration 2–4 in Section 2 for required fork dimensions. The minimum fork length is 2.1 Meters (90 inches).
- S The magnet pallet MUST be picked up from the sides only, with the forklift forks squarely under all four (4) feet of the magnet
- S The magnet MUST be lifted smoothly to avoid impact or jolts to the system which may cause damage to the magnet.

1. Position the forklift forks at the forklift openings in the side of the shipping pallet. See Illustrations 3-7 and 3-8.
2. Carefully drive the forklift until the forks are completely inserted into the pallet openings.
3. Lift the magnet / pallet with the forklift and smoothly move to the desired location.



FORKLIFT / CRANE LIFTING OPERATIONS
ILLUSTRATION 3-7

3-4-1 Forklift Method (continued)

LIFTING CONFIGURATIONS FOR FORKLIFT
ILLUSTRATION 3-8

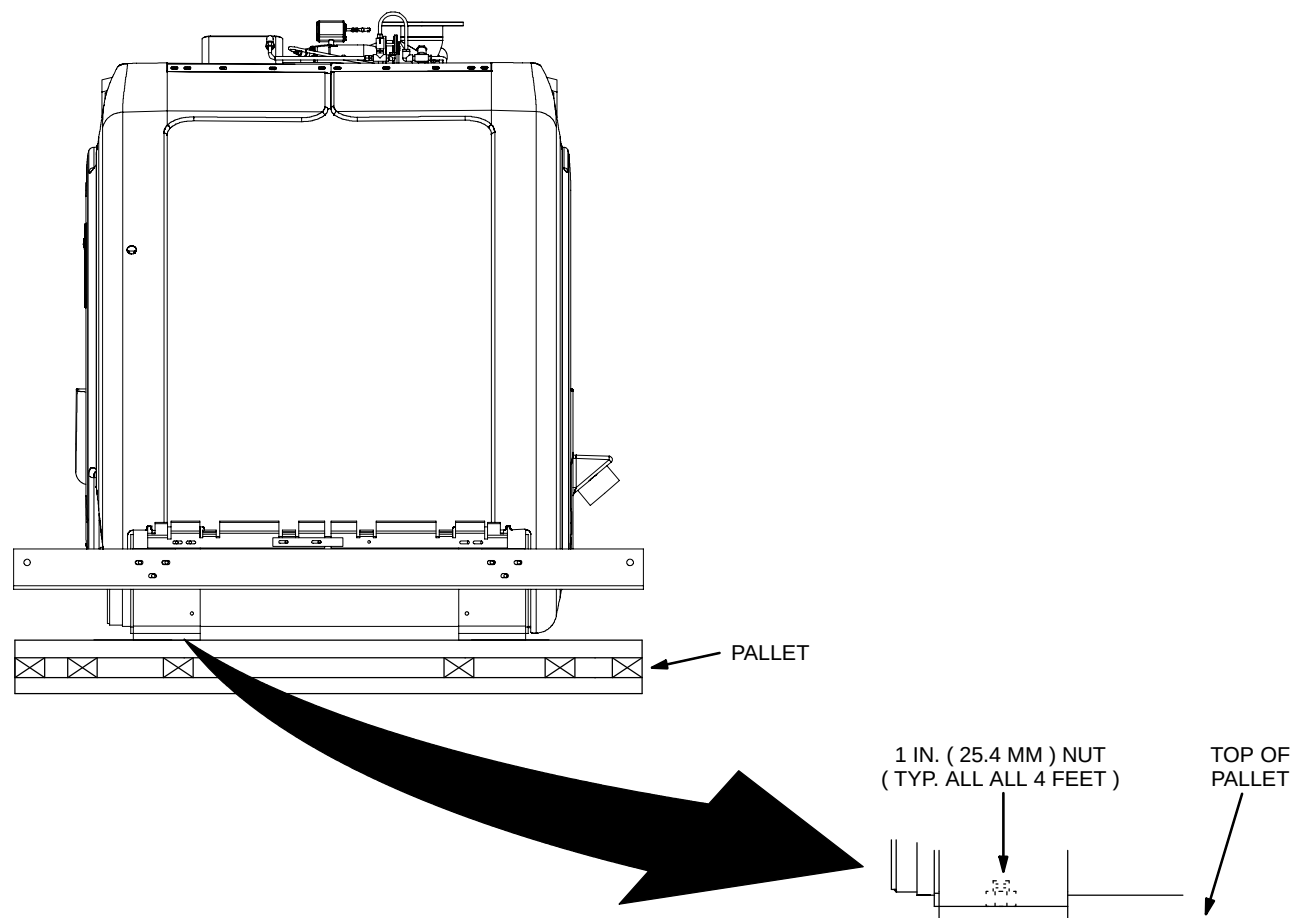
3-4-2 Crane Method



The pallet **MUST** be removed from under the magnet prior to any magnet lift operations.

1. Unbolt and remove the four 1 inch (25.4 mm) nuts which secure the pallet to the magnet. See Illustration 3-9.
2. For rigging and lifting details, refer to Section 2.

3-4-2 Crane Method (continued)

MAGNET MOUNTING ON SHIPPING PALLET
ILLUSTRATION 3-9

